

A new consensus-based formation tracking scheme for a class of robotic systems using negative imaginary property

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Abstract: This brief proposes a consensus-based formation tracking scheme for multi-robot systems utilising the Negative Imaginary (NI) theory. The proposed scheme applies to a class of networked robotic systems that can be modelled as a group of single integrator agents with stable uncertainties connected via an undirected graph. NI/SNI property of networked agents facilitates the design of a distributed Strictly Negative Imaginary (SNI) controller to achieve the desired formation tracking. A new theoretical proof of asymptotic convergence of the formation tracking trajectories is derived based on the integral controllability of networked NI/SNI systems. The proposed scheme is an alternative to the conventional Lyapunov-based formation tracking schemes. It offers robustness to NI/SNI-type model uncertainties and fault-tolerance to a sudden loss of robots due to hardware/communication fault. The feasibility and usefulness of the proposed formation tracking scheme were validated by lab-based real-time hardware experiments involving miniature, two-wheeled mobile robots.

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1. INTRODUCTION

Negative Imaginary (NI) systems theory was introduced in [Lanzon and Petersen (2008)] and has gradually come into the limelight due to its potential applications vibration control of lightly damped flexible structures [Lanzon and Petersen (2008)], cantilever beams [Bhikkaji et al. (2012)], large space structures [Mabrok et al. (2014)], robotic manipulators [Mabrok et al. (2014)], control of nano-positioning systems [Das et al. (2014)], control of multi-agent networked systems [Tran et al. (2020); Wang et al. (2015); Wang et al. (2015)], etc. Simplistically, the term NI refers to a class of LTI systems having negative imaginary frequency response, that is, the imaginary part of an NI transfer function remains non-positive $\forall \omega \in (0, \infty)$; while that of a Strictly Negative Imaginary (SNI) system transfer function remains negative $\forall \omega \in (0, \infty)$. It gives an interesting graphical interpretation for NI and SNI systems in the SISO setting – the Nyquist plot of an NI (resp. SNI) transfer function lies below (resp. strictly below) the real-axis in the complex plane $\forall \omega \in (0, \infty)$ [Lanzon and Petersen (2008)]. Negative imaginary framework offers a simple closed-loop stability criterion that depends only on the steady-state gains of the systems and hence, it becomes useful for designing controllers for practical systems even when the exact mathematical model is not available

[Bhowmick and Lanzon (2019, 2020, 2021); Bhowmick and Patra (2017, 2020)].

Cooperative control of multi-robot systems has been an active area of research to the robotics and control communities over the past two decades. Some of the potential applications of the cooperative control strategies include cooperative exploration, object transportation [Hu et al. (2021b)], vehicle platooning [Hu et al. (2020a)], etc. The primary motivation of applying NI systems theory to develop a leader-following formation control scheme for networked multi-robot systems is that many robotic systems (e.g. wheeled mobile robot with nonholonomic constraints [Hu et al. (2021a)]) can be transformed into single or double integrator dynamics via input-output feedback linearization [Slotine and Li (1991)] and furthermore, a network of single/double integrator dynamics, being a MIMO NI system, can easily be stabilized by a MIMO SNI systems in a positive feedback loop [Wang et al. (2015); Wang et al. (2015)]. Motivated by the aforementioned applications, this paper aims to develop a simple leader-following consensus control scheme for networked single integrator agents. This technical note introduces a new methodology (instead of the commonly used Lyapunov-based approach) to prove convergence of the leader-following consensus control problem, which encompasses (static) formation control and cooperative tracking as particular cases, exploiting the

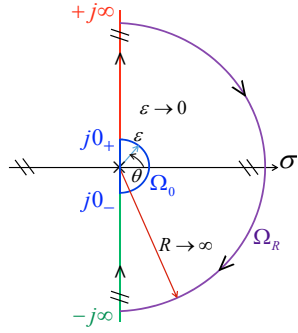


Fig. 1. Nyquist $\mathcal{D}$ -contour in the s -plane.

characteristics of the eigenvalue loci of networked NI and SNI systems.

2. ESSENTIAL PRELIMINARIES

In this section, some valuable technical preliminaries, definitions and lemmas are presented which underpin the proofs of the main results of the brief.

2.1 Definitions for negative imaginary systems theory

In this subsection, we recall the definitions of NI and SNI systems.

Definition 1. (NI System) [Mabrok et al. (2014)] Let $G(s)$ be the real, rational, proper transfer function matrix of a finite-dimensional and square system without any poles in $\Re[s] > 0$. Then, $G(s)$ is said to be NI if

- $j[G(j\omega) - G(j\omega)^*] \geq 0$ for all $\omega \in (0, \infty)$ except the values of ω where $j\omega$ is a pole of $G(s)$;
- If $s = j\omega_0$ with $\omega_0 \in (0, \infty)$ is a pole of $G(s)$, then it is at most a simple pole and the residue matrix $\lim_{s \rightarrow j\omega_0} (s - j\omega_0)jG(s)$ is Hermitian and positive semidefinite;
- If $s = 0$ is a pole of $G(s)$, then $\lim_{s \rightarrow 0} s^k G(s) = 0$ for all $k \geq 3$ and $\lim_{s \rightarrow 0} s^2 G(s)$ is Hermitian and positive semidefinite.

Definition 2. (SNI System) [Lanzon and Petersen (2008)] Let $G(s)$ be the real, rational, proper transfer function matrix of a finite-dimensional and square system. Then, $G(s)$ is said to be SNI if $G(s)$ has no poles in $\Re[s] \geq 0$ and $j[G(j\omega) - G(j\omega)^*] > 0$ for all $\omega \in (0, \infty)$.

2.2 Eigenvalue loci theory

Like the Nyquist plot, the eigenvalue loci $\gamma_i(s)$ for $i \in \{1, 2, \dots, m\}$ of a transfer function matrix $P(s)$ is a conformal mapping of the function $\det[P(s)]$ in a complex plane, known as the eigenvalue loci plane, when s traverses along the s -plane D -contour in the clockwise direction as shown in Fig. 1. For complete details of the eigenvalue loci theory, the literature [Belletrutti and MacFarlane (1971); MacFarlane and Belletrutti (1973)] may be referred.

Theorem 3. (Belletrutti and MacFarlane (1971)). The negative feedback interconnection of a plant $G(s)$ and a controller $C(s)$ is stable if, and only if, the net sum of critical point $(-1 + j0)$ encirclements of all the eigenvalue loci

$\gamma_i(j\omega)$ of the loop transfer function $G(s)C(s)$ for $i \in \{1, 2, \dots, m\}$ is counter-clockwise and equal to the number of RHP zeros of the open-loop characteristic polynomial. For open-loop stable cases, none of $\gamma_i(j\omega)$ should encircle the critical point in the complex plane.

2.3 Graph theory notations

Consider a weighted and directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$ with a non-empty set of nodes $\mathcal{V} = \{1, 2, \dots, M\}$, a set of edges $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$, and the associated adjacency matrix $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{M \times M}$. An edge rooted at the i^{th} node and ended at the j^{th} node is denoted by (i, j) , which means that the information can flow from node i to node j . a_{ij} is the weight of the edge (j, i) and $a_{ij} > 0$ if $(j, i) \in \mathcal{E}$. The j^{th} node is called a neighbour of the i^{th} node if $(j, i) \in \mathcal{E}$. The in-degree matrix is defined as $\mathcal{D} = \text{diag}\{d_i\} \in \mathbb{R}^{M \times M}$ where $d_i = \sum_{j=1}^M a_{ij}$. The Laplacian matrix $L \in \mathbb{R}^{M \times M}$ of $\mathcal{G}$ is defined as $L = \mathcal{D} - \mathcal{A}$.

Consider a multi-robot system containing M robots and one target. The target can be observed by a subset of robots in the network. If the i^{th} robot observes the target, an edge $(0, i)$ is said to exist between them with a pinning gain $p_i > 0$.

3. NETWORKED NI AND SNI SYSTEMS

In this section, we will first discuss under which conditions a network of stable NI or SNI systems retains the same property. Subsequently, we will explore the behaviour of eigenvalue loci of networked NI and SNI systems. Finally, the integral controllability property of networked NI and SNI systems has been investigated in the homogeneous case. These results are essential prerequisites which will be invoked in Section 4 to develop a simple distributed formation control scheme for networked single integrator systems applying NI toolkit.

3.1 Properties of networked NI and SNI systems

We will now declare two technical assumptions to be satisfied by the communication graphs corresponding to the network of NI/SNI systems considered in this brief.

Assumption 1. The interaction among M agents is described by an undirected and connected graph $\mathcal{G}$. There always exists a root node which represents a leader (or a virtual target) and it provides the reference trajectory to the follower agents (at least to one follower).

According to Assumption 1, we have $L + P > 0$ where $P = \text{diag}\{p_1, p_2, \dots, p_M\} > 0$ is the pinning-gain matrix. We will use the shorthand $\mathcal{L}_M = L + P$ throughout this brief. The following lemma proves that a network of homogeneous LTI systems exhibits stable NI (SNI) property if and only if an unit system is stable NI (SNI).

Lemma 4. Consider a group of M stable NI (SNI) agents $G(s) \in \mathcal{RH}_\infty^{m \times m}$ connected via a network topology $\mathcal{G}$ satisfying Assumption 1. Then, $\bar{G}(s) = \mathcal{L}_M \otimes G(s)$ is stable NI (SNI) if and only if $G(s) \in \mathcal{RH}_\infty^{m \times m}$ is NI (SNI).

Proof. (Sufficiency) Since $G(s)$ is stable NI (SNI), it satisfies $j[G(j\omega) - G(j\omega)^*] \geq 0$ (> 0) $\forall \omega \in (0, \infty)$. Now

we have $j[\bar{G}(j\omega) - \bar{G}(j\omega)^*] = j[\mathcal{L}_M \otimes G(j\omega) - \mathcal{L}_M^\top \otimes G(j\omega)^*] = \mathcal{L}_M \otimes j[G(j\omega) - G(j\omega)^*]$ [since $\mathcal{L}_M = \mathcal{L}_M^\top > 0$] ≥ 0 (> 0) $\forall \omega \in (0, \infty)$ by applying the Kronecker product property $A \otimes B \geq 0$ when $A = A^* \geq 0$ and $B = B^* > 0$ [Horn and Johnson (2012)].

(Necessity) $j[\bar{G}(j\omega) - \bar{G}(j\omega)^*] \geq 0$ (> 0) $\forall \omega \in (0, \infty)$ implies $j[G(j\omega) - G(j\omega)^*] \geq 0$ (> 0) $\forall \omega \in (0, \infty)$ since $\mathcal{L}_M = \mathcal{L}_M^\top > 0$ due to satisfying Assumption 1. This completes the proof. ■

Lemma 2 shows that a homogeneous network of stable NI (including SNI) systems has the same DC-gain matrix property as that of its unit subsystem when the corresponding graph satisfies Assumption 1

Lemma 5. Consider a group of M stable NI (SNI) agents $G(s) \in \mathcal{RH}_\infty^{m \times m}$ connected via a network topology $\mathcal{G}$ satisfying Assumption 1. Denote $\bar{G}(s) = \mathcal{L}_M \otimes G(s)$. Then, $\bar{G}(0) > 0$ (< 0) if and only if $G(0) > 0$ (< 0).

Proof. (Sufficiency) Since $\mathcal{L}_M \otimes G(0) = \mathcal{L}_M^\top \otimes G(0)^\top$ and $\lambda_i[\mathcal{L}_M \otimes G(0)] = \lambda_j[\mathcal{L}_M] \lambda_k[G(0)]$ when $G(0) > 0$ (< 0) by exploiting the eigenvalue property of a Kronecker product [Horn and Johnson (2012)] with $i \in \{1, 2, \dots, Mm\}$, $j \in \{1, 2, \dots, M\}$ and $k \in \{1, 2, \dots, m\}$, it follows $\bar{G}(0) = \mathcal{L}_M \otimes G(0) > 0$ (< 0). The symbol $\lambda[\cdot]$ denotes the eigenvalues of a square matrix.

The Necessity part of the proof follows from the same technical arguments used in the proof of Lemma 4. ■

3.2 Properties of eigenvalue loci of networked NI systems

So far in the literature, the eigenvalue loci theory [Belletrutti and MacFarlane (1971); MacFarlane and Belletrutti (1973)] has been defined for a single LTI system. In this brief, being inspired by the developments of [Bhowmick and Patra (2018)], eigenvalue loci technique has been extended to networked NI and SNI systems in the homogeneous case. The following lemma reveals that the eigenvalue loci of an homogeneous network of stable NI (SNI) systems stay within the third and fourth quadrants of a complex plane, termed as the eigenvalue loci plane, including (excluding) the real-axis in the open positive frequency interval, i.e., $\forall \omega \in (0, \infty)$.

Lemma 6. Consider a group of M stable NI (SNI) agents $G(s) \in \mathcal{RH}_\infty^{m \times m}$ connected via a network topology $\mathcal{G}$ satisfying Assumption 1. Denote $\bar{G}(s) = \mathcal{L}_M \otimes G(s)$. Then, the eigenvalue loci $\gamma_i(j\omega)$ of $\bar{G}(s) \in \mathcal{RH}_\infty^{Mm \times Mm}$ lie below (strictly below) the real-axis of the eigenvalue loci plane $\forall \omega \in (0, \infty)$ and $\forall i \in \{1, 2, \dots, Mm\}$.

Proof. We start this proof by recalling that $\bar{G}(s) = \mathcal{L}_M \otimes G(s)$ is stable NI (SNI) if and only if $G(s)$ is stable NI (SNI) via Lemma 4. Assume (λ_i, x_i) be an eigenvalue-eigenvector pair for $(\mathcal{L}_M \otimes G(j\omega)) \in \mathbb{C}^{Mm \times Mm}$ for $\omega \in (0, \infty)$ where $\lambda_i \in \mathbb{C}$ for $i \in \{1, 2, \dots, Mm\}$ and $0 \neq x_i \in \mathbb{C}^{Mm}$. Note that in the case of repeated eigenvalues of $(\mathcal{L}_M \otimes G(j\omega))$ at an $\omega \in (0, \infty)$, only the linearly independent eigenvectors needs to be considered. Now,

$$\begin{aligned} & x_i^* (\mathcal{L}_M \otimes G(j\omega)) x_i \\ &= x_i^* [\mathcal{L}_M \otimes [\frac{1}{2}(G(j\omega) + G^*(j\omega)) + j\frac{1}{2j}(G(j\omega) - G^*(j\omega))]] x_i \\ &= x_i^* (\mathcal{L}_M \otimes A) x_i + j x_i^* (\mathcal{L}_M \otimes B) x_i \end{aligned} \quad (1)$$

where $A = \frac{1}{2}[G(j\omega) + G^*(j\omega)]$ and $B = \frac{1}{2j}[G(j\omega) - G^*(j\omega)]$. Considering an orthonormal eigenvector $\|x_i\|_2 = 1$, without loss of generality, we obtain $x_i^* (\mathcal{L}_M \otimes G(j\omega)) x_i = \lambda_i(x_i^* x_i) = \lambda_i \|x_i\|_2 = \Re[\lambda_i] + j \Im[\lambda_i]$. (2)

Upon comparing (1) and (2), we have $\Re[\lambda_i] = x_i^* (\mathcal{L}_M \otimes A) x_i$ and $\Im[\lambda_i] = x_i^* (\mathcal{L}_M \otimes B) x_i$, both of which are real and scalar quantities since $A = A^*$, $B = B^*$ and $\mathcal{L}_M = \mathcal{L}_M^\top > 0$. Now, $B \leq 0$ (< 0) $\forall \omega \in (0, \infty)$ since $G(s)$ is stable NI (SNI) using Definitions 1 and 2. This then implies $\lambda_i[\mathcal{L}_M \otimes B] = \lambda_j[\mathcal{L}_M] \lambda_k[B] \leq 0$ (< 0) $\forall \omega \in (0, \infty)$ and for all i via exploiting the eigenvalue property of the Kronecker product of two matrices [Horn and Johnson (2012)] when $G(s)$ is stable NI (SNI). Therefore, for stable NI (SNI) systems we have $\Im[\lambda_i\{\mathcal{L}_M \otimes G(j\omega)\}] \leq 0$ (< 0) $\forall \omega \in (0, \infty)$ and $\forall i \in \{1, 2, \dots, Mm\}$. This is hence a proof that all $\gamma_i(j\omega)$ of the network of homogeneous stable NI (SNI) systems lie below (strictly below) the real-axis of the eigenvalue loci plane $\forall \omega \in (0, \infty)$. ■

Note that although the eigenvalue loci of the SNI systems $G(s) \in \mathcal{RH}_\infty^{m \times m}$ do not touch the real axis of the eigenvalue loci plane for any $\omega \in (0, \infty)$, but each locus originates from (at $\omega = 0$) and terminate (at $\omega = \infty$) to the real axis since $G(0) = G(0)^\top$ and $G(\infty) = G(\infty)^\top$.

4. A FORMATION TRACKING SCHEME FOR ROBOTIC AGENTS USING NI THEORY

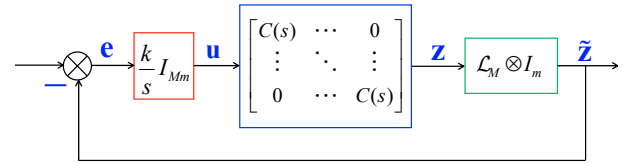


Fig. 2. An integrator feedback control scheme for networked stable NI systems connected via the graph $\mathcal{G}$.

This section presents the main contributions of the paper. A simple but effective formation tracking scheme has been developed for a class of networked robotic systems that can be modelled or transformed into a group of single integrator agents, utilising the NI theory. The following lemma is given first which is an essential technical prerequisite to prove the main result (i.e. Theorem 8).

Lemma 7. (Hu et al. (2020c)). Consider a group of M stable NI systems $C(s) \in \mathcal{RH}_\infty^{m \times m}$ with $C(0) > 0$ connected via a network topology $\mathcal{G}$ that satisfies Assumption 1. Then, there exists a finite $k^* > 0$ such that the negative feedback interconnection of $(\mathcal{L}_M \otimes C(s))$ and $(I_{Mm} \otimes \frac{k}{s})$, in Fig. 2, remains asymptotically stable for all $k \in (0, k^*]$.

In the context of cooperative control of multi-robot systems, serious problems occur due to a sudden loss of agents, resulting from hardware faults or network failure. In such situations it is of interest to determine how to preserve the stability of a network. Being motivated by the developments of [Bhowmick and Patra (2018)] and [Hu et al. (2020c)], in this section, we will establish that the proposed formation tracking scheme for networked

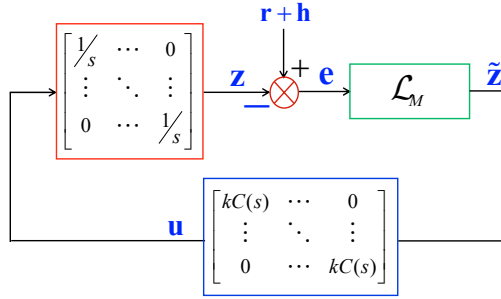


Fig. 3. A formation tracking scheme for a group of networked single integrator agents applying NI theory.

single integrator systems facilitate failure tolerance. That is, upon occurrence of a fault the control scheme maintains the overall stability of the network and also, a new stable operating condition can easily be reached subjected to an autonomous reconfiguration of the network excluding the faulty agents. In the control scheme, the loss of an agent is modelled by making the gain of that particular channel equal to zero which means that the faulty loop is temporarily deactivated.

Theorem 8, given next, will establish that, under the application of a MIMO, decoupled stable NI controller $\text{diag}\{C(s), \dots, C(s)\}$ where $C(s)$ is a stable NI transfer function with $C(0) > 0$, a group of single integrator agents connected via a graph topology $\mathcal{G}$ satisfying Assumption 1 will asymptotically reach the desired formation and continue to track the leader.

Theorem 8. Let M single integrator agents be connected via the topology $\mathcal{G}$, which satisfies Assumption 1 and the set of admissible reference inputs $\mathbf{r} \in \mathcal{L}_\infty^M$. Let $\mathbf{h} = [h_1 \ h_2 \ \dots \ h_M]^\top \in \mathbb{R}^M$ be the desired formation configuration vector. Let $C(s)$ be a stable NI transfer function with $C(0) > 0$. Then, there exists a finite $k^* > 0$ such that the single integrator agents in Fig. 3 achieve consensus-based formation tracking by the following distributed stable NI output feedback control law

$$u_i = k_i C(s) \sum_{j=1}^M a_{ij} ((z_i - h_i) - (z_j - h_j)) + p_i (z_i - h_i - r) \quad (3)$$

for any $k_i \in [0, k^*]$ and for each $i \in \{1, 2, \dots, M\}$.

Proof. In this proof, the notations $\mathbf{E}(s)$, $\mathbf{Z}(s)$, $\mathbf{R}(s)$ will represent the Laplace transform of the real-valued, time-domain vector signals $\mathbf{e}(t) = [e_1(t) \ e_2(t) \ \dots \ e_M(t)]^\top$, $\mathbf{z}(t) = [z_1(t) \ z_2(t) \ \dots \ z_M(t)]^\top$ and $\mathbf{r} \in \mathcal{L}_\infty^M$. Note that the controller $C(s)$ is a stable NI transfer function with $C(0) > 0$.

Case I: when $k_i \in (0, k^*] \ \forall i \in \{1, 2, \dots, M\}$

Let $k \in [\inf_i k_i, \sup_i k_i]$ where $0 < \inf_i k_i < \sup_i k_i$.

Part 1: It can be readily ensured via Lemma 7 that the negative feedback single integrator consensus scheme shown in Fig. 3 is asymptotically stable under the application of the distributed stable NI control law u_i for all $k \in [\inf_i k_i, \sup_i k_i]$. This implies that the closed-loop stability of the network holds for all $k \in (0, k^*]$.

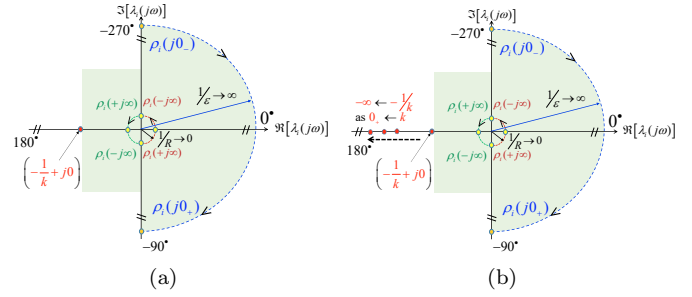


Fig. 4. (a) All the eigenvalue loci $\gamma_i(j\omega)$ of $\mathcal{L}_M \otimes \frac{1}{s}C(s)$ remain confined within the Yellow coloured region $\forall \omega \in \mathbb{R} \cup \{\infty\}$ when $C(s)$ is a stable NI transfer function having $C(0) > 0$; and (b) None of the eigenvalue loci $\gamma_i(j\omega)$ of $\mathcal{L}_M \otimes \frac{1}{s}C(s)$ does encircle the critical points $(-\frac{1}{k_i} + j0)$ for any $k_i \in [0, k^*]$ and for each $i \in \{1, 2, \dots, M\}$.

Part 2: With respect to Fig. 3, we define the formation trajectory tracking error as $\mathbf{e}(t) \triangleq \mathbf{r}(t) + \mathbf{h} - \mathbf{z}(t)$ and denote $\hat{\mathbf{r}} \triangleq \mathbf{r} + \mathbf{h}$. Now, we have

$$\mathbf{E}(s) = \left[I + \left(\mathcal{L}_M \otimes \frac{k}{s} C(s) \right) \right]^{-1} \hat{\mathbf{R}}(s). \quad (4)$$

The steady-state formation tracking error is found to be

$$\mathbf{e}_{ss} = \lim_{t \rightarrow \infty} \mathbf{e}(t) = \lim_{s \rightarrow 0} s \mathbf{E}(s) \quad (5a)$$

$$\begin{aligned} &= \lim_{s \rightarrow 0} s \left[I_M + \left(\mathcal{L}_M \otimes \frac{k}{s} C(s) \right) \right]^{-1} \hat{\mathbf{R}}(s) \\ &= \lim_{s \rightarrow 0} s [s I_M + (\mathcal{L}_M \otimes k C(s))]^{-1} (s \hat{\mathbf{R}}(s)) \\ &= [\mathcal{L}_M \otimes (k C(0))]^{-1} \left(\lim_{s \rightarrow 0} s I_M \right) \left(\lim_{s \rightarrow 0} s \left[\mathbf{R}(s) + \frac{1}{s} \mathbf{h} \right] \right) \\ &= [0 \ 0 \ \dots \ 0]^\top \end{aligned} \quad (5b)$$

since $C(0) > 0$, $\mathcal{L}_M > 0$ and $\lim_{t \rightarrow \infty} \mathbf{r}(t) \triangleq \mathbf{r}_{ss}$ exists and is finite via assumption. The asymptotic convergence property of the trajectory tracking error, that is $\lim_{t \rightarrow \infty} \mathbf{e}(t) = \mathbf{0}$ implies $\mathbf{z}(t) \rightarrow \hat{\mathbf{r}}_{ss} = (\mathbf{r}_{ss} + \mathbf{h})$ as $t \rightarrow \infty$.

Case II: when $k_i \in [0, k^*] \ \forall i \in \{1, 2, \dots, M\}$

From **Case I**, asymptotic stability of the formation tracking scheme for networked single integrator agents in Fig. 3 is guaranteed by the distributed stable NI control law (3) for any $k_i \in (0, k^*]$ and for each $i \in \{1, 2, \dots, M\}$. This is equivalent to fulfilling the requirement that none of the eigenvalue loci $\gamma_i(s)$ of $\mathcal{L}_M \otimes \frac{1}{s}C(s)$ encircles the critical point $(-\frac{1}{k_i} + j0)$ for any value of $k_i \in (0, k^*]$. When $k_i = 0$ for some $i \in \{1, 2, \dots, M\}$, the critical point $(-\frac{1}{k_i} + j0)$ reaches $(-\infty + j0)$ as depicted in Fig. 4(b). According to Theorem 8, all the eigenvalue loci $\gamma_i(j\omega)$ remain within the Yellow coloured region in Fig. 4(b) and therefore, no infinite crossover takes place on the negative real-axis of the eigenvalue loci plane. This hence ensures that none of the critical points $(-\frac{1}{k_i} + j0)$ where $k_i \in [0, k^*]$ will be encircled by the eigenvalue loci $\gamma_i(j\omega)$. This implies that closed-loop stability remains preserved even when some of the input-output channels are broken (indicated by $k_i = 0$). This completes the proof.

5. EXPERIMENTAL VALIDATION

To examine the feasibility and effectiveness of the proposed NI-based formation control scheme, lab-based hardware experiments involving small-scale two-wheeled mobile robots (Mona [Arvin et al. (2019)] shown in Fig. 5(a)) were conducted. The hardware control loop is shown in Fig. 5(b) along with the experimental set-up which utilises a camera tracking system and a host computer. The position tracking system used in this experiment is an open-source multi-robotic localization system [Krajník et al. (2014)]. The tracking system can track the positions and orientations of the robots by identifying the unique circular tags attached on top of the robots. The position information is transmitted to the controller via the ROS communication framework. An input-output feedback linearization technique was applied to transform the nonlinear kinematics of the two-wheeled mobile robots (Mona) into a decoupled two-input-two-output single integrator system ($\frac{1}{s}I_2$) following [Hu et al. (2020b)]. In order to avoid possible collisions among the robots during their movement in the arena, an Artificial Potential Field (APF) algorithm [Tnunay et al. (2017)] was implemented through the low-level controller acting on the mobile robots. Each robot had a safe distance of 0.1 m and was repelled from other robots if they were within its safe distance.

5.1 Experiment

This experiment was performed with four Mona robots under the application of a distributed SISO stable NI controller $C(s) = \frac{2s^2+s+1}{(s+1)(2s+1)(s^2+2s+5)}$ with $C(0) > 0$ to test the robustness of the proposed formation tracking scheme to sudden loss of agents. The experiment started with the objective of achieving a square formation by the robots with respect to the given virtual target (marked by ●). Fig. 6(a) shows the initial configuration of the robots on the arena, while Fig. 6(b) suggests that the desired square formation was attained by those four robots surrounding the virtual target. Fig. 6(c) shows that the robots' formation as a whole kept on moving towards the centre of the arena as the virtual target was also moving. Fig. 6(c) gave rise to a situation when one of the robots (marked by X) stopped functioning due to a sudden fault. Fig. 6(d) shows that despite a sudden loss of one robot, the remaining three were able to tackle the situation and attained a new triangular formation with respect to the given target. Please note here that the switching from square formation to triangular formation was made possible via an autonomous network reconfiguration (implemented via a Matlab program) of the existing robots when one of them stopped operating due to a fault. During the transient phase, upon occurrence of the fault, the overall stability of the network is guaranteed via Theorem 8. Fig. 7 complements the results depicted in Fig. 6(a)–6(d) by showing the spatial variation of the position trajectories of the robots during the experiment.

6. CONCLUSIONS

This brief has developed a consensus-based formation tracking scheme for a class of multi-robot system that can be modelled as (or feedback linearized into) a network

of single integrator agents. Owing to the NI property of networked agents connected via an undirected graph, a distributed stable NI controller facilitates a formation tracking scheme with asymptotic convergence. The eigenvalue loci technique is used as an alternative to the conventional Lyapunov-based approaches to derive a theoretical proof of the proposed scheme. Lab-based experiments involving small-scale two-wheeled mobile robots were performed to show the feasibility of the scheme.

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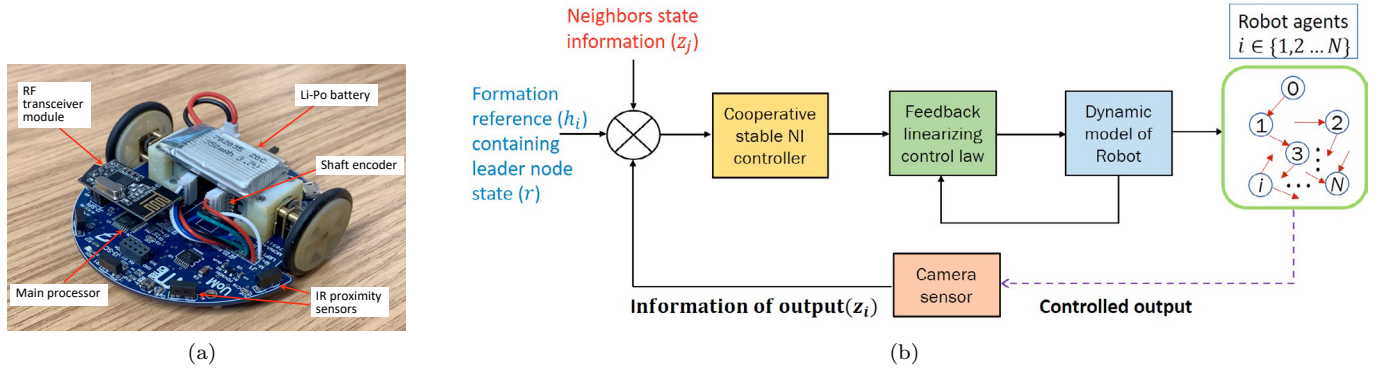


Fig. 5. (a) Mona robot [Arvin et al. (2019)] used in the experimental validation; and (b) The hardware control loop implemented in our experiment. The experimental set-up also includes a camera tracking system and a host computer.

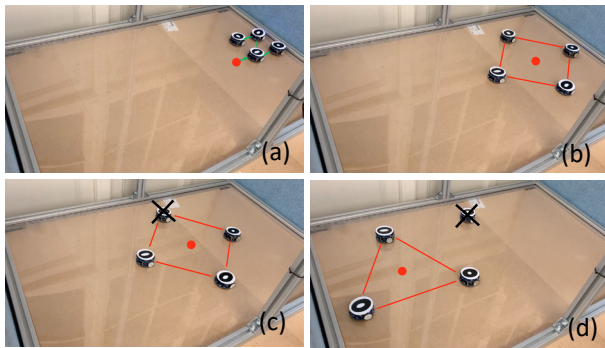


Fig. 6. Experiment 2 shows that a square formation is first attained by four Mona Robots, which is then switched to a triangular formation after a sudden loss of one robot.

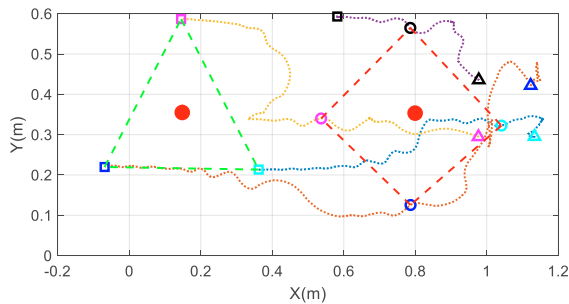


Fig. 7. Position trajectories of the robots during the formation tracking mission in Experiment 2.

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